NETWORK ENGINEERING · FOUNDATIONS

IPV4: THE PROTOCOL THAT RUNS THE INTERNET

From packet headers to practical subnet design

Read addresses in binary. Calculate subnet boundaries. Recognize the ranges that never belong on the public Internet.

`IPv4 = Internet Protocol version 4`



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PROTOCOL ROLE

WHAT IPV4 IS AND WHERE IT SITS

IPv4 is a connectionless, best-effort Layer 3 protocol defined in RFC 791 (1981). Each datagram is handled independently; delivery, ordering and duplicate protection are not guaranteed.

Application

HTTP, DNS and other application protocols

Transport

TCP or UDP; end-to-end ports

Internet · OSI Layer 3

IPv4; logical addresses and routed packets

Link · OSI Layers 1–2

Ethernet or Wi-Fi; local frame delivery



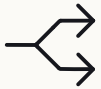
Addressing

Identifies source and destination interfaces with 32-bit addresses.



Routing

Routers forward packets toward the destination network.



Fragmentation

Fits packets to a link’s maximum transmission unit when permitted; the destination reassembles them.

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EXHAUSTION

IPV4’S FREE POOL RAN OUT; EXISTING ADDRESSES KEPT WORKING.

Regional Internet Registries (RIRs) have different reserve, transfer and recovery policies.

EVOLUTION

A SHORT HISTORY OF IPV4

IPv4 grew from ARPANET-era internetworking: a common packet format let different underlying networks communicate.

1981	RFC 791 defines IPv4, including its 32-bit address space and datagram header.
1983	The January 1 ARPANET “flag day” switches hosts from NCP to TCP/IP.
1993	Classless Inter-Domain Routing (CIDR) replaces rigid class boundaries with variable prefixes.
2011 onward	IANA exhausts its unallocated IPv4 pool; regional registries reach exhaustion milestones afterward.

Classful allocation → CIDR → address conservation and IPv6

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ADDRESS REPRESENTATION

ANATOMY OF AN IPV4 ADDRESS

An IPv4 address contains 32 bits in four 8-bit octets. Dotted decimal writes each octet as a value from 0 to 255.

192.168.10.34/24 · four octets · 32 bits

192	168	10	34
11000000	10101000	00001010	00100010

The prefix sets the boundary

With /24, the first 24 bits identify network 192.168.10.0. The remaining 8 bits identify the host within that subnet. The address alone does not reveal this split.

Read binary by place value

From left to right, bit weights are 128, 64, 32, 16, 8, 4, 2, 1. Add only weights whose bits are 1.

$$11000000_2 = 128 + 64 = 192_{10}$$

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PACKET FORMAT · EACH ROW IS 32 BITS

THE IPV4 HEADER, FIELD BY FIELD

Version 4 bits	IHL 4 bits	DSCP 6 bits	ECN 2	Total Length · 16 bits	
Identification · 16 bits				Flags 3	Fragment Offset · 13 bits
TTL · 8 bits		Protocol · 8 bits		Header Checksum · 16 bits	
Source Address · 32 bits					
Destination Address · 32 bits					

Minimum header: 20 bytes. Optional Options and padding follow the destination address; maximum header: 60 bytes.

Version 4; IHL counts 32-bit words (minimum 5). DSCP marks traffic treatment; ECN signals congestion. Total Length includes header and payload, up to 65,535 bytes. Identification groups fragments; Flags include Don’t Fragment (DF) and More Fragments (MF); Fragment Offset uses 8-byte units.

Time to Live (TTL) decreases at each router. Protocol identifies the payload: ICMP 1, TCP 6, UDP 17. Header Checksum protects only the header and is updated when it changes. Source and Destination carry endpoint addresses; Options are variable-length and uncommon.

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HISTORICAL ADDRESSING MODEL

CLASSFUL ADDRESSING: CLASSES A TO E

The leading bits selected a class and, for unicast Classes A–C, an implied network mask.

Class / bits / prefix	Full address range	Original role
A · 0 · /8	0.0.0.0–127.255.255.255	Large unicast networks
B · 10 · /16	128.0.0.0–191.255.255.255	Medium unicast networks
C · 110 · /24	192.0.0.0–223.255.255.255	Small unicast networks
D · 1110 · None	224.0.0.0–239.255.255.255	Multicast groups
E · 1111 · None	240.0.0.0–255.255.255.255	Experimental / reserved

Ranges show class boundaries, not general assignability. For example, 0/8 and 127/8 are special-purpose; 255.255.255.255 is the limited broadcast address.

Modern subnets use configured prefixes, not classes; CIDR lets 10.x.x.x use /24, /26 or another valid prefix

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THE NETWORK / HOST BOUNDARY

SUBNET MASKS AND CIDR NOTATION

Masks place contiguous 1 bits before 0 bits. CIDR /n counts network bits; the rest are host bits.

Subnet mask · /26

`255.255.255.192`

11111111.11111111.
11111111.11000000

Bitwise AND of an address and the mask gives its network address.

Wildcard mask

`0.0.0.63`

The inverse mask is used in access control lists (ACLs) and OSPF network statements.

0 means “match this bit”; 1 means “ignore this bit.” Wildcards need not be contiguous.

Count usable hosts

$$h = 32 - n, \qquad H = 2^h - 2$$

For /26: 6 host bits, 64 addresses, and 62 conventional usable host addresses.

Know the exceptions

Ordinary subnets reserve network and broadcast addresses. A /31 addresses both ends of a point-to-point link; a /32 identifies one address.

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WORKED EXAMPLE · 192.168.10.0/24 → /26

SUBNETTING STEP BY STEP

01

Pick the prefix

For up to 62 hosts per subnet, retain 6 host bits. The prefix is /26.

02

Find the increment

Mask 255.255.255.192. Last-octet block size is 256 − 192 = 64.

03

Mark the boundaries

Networks start at .0, .64, .128 and .192. Each broadcast is one below the next boundary.

Borrow 2 bits · 4 equal subnets · 64 addresses each

Network / prefix	Usable host range	Broadcast	Hosts
192.168.10.0/26	192.168.10.1–192.168.10.62	192.168.10.63	62
192.168.10.64/26	192.168.10.65–192.168.10.126	192.168.10.127	62
192.168.10.128/26	192.168.10.129–192.168.10.190	192.168.10.191	62
192.168.10.192/26	192.168.10.193–192.168.10.254	192.168.10.255	62

Check a host by locating its block: 192.168.10.130/26 belongs to 192.168.10.128/26, not 192.168.10.0/26.

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ADDRESS PLANNING AND ROUTE CONTROL

VLSM AND ROUTE SUMMARIZATION

Variable-Length Subnet Masking (VLSM) uses different prefix lengths within one address block. Allocate the largest requirements first, preserving alignment and avoiding overlap.

Divide to fit the demand

Illustrative allocation from 192.168.10.0/24:

100 hosts → 192.168.10.0/25 — 126 usable; covers .0–.127

50 hosts → 192.168.10.128/26 — 62 usable; covers .128–.191

20 hosts → 192.168.10.192/27 — 30 usable; covers .192–.223

192.168.10.224/27 remains available.

Combine to simplify routing

192.168.10.0/25 + 192.168.10.128/25 →

192.168.10.0/24

Contiguous, aligned blocks sharing the first 24 bits combine into one advertised route.

Summaries reduce routing-table size and can contain route-change propagation. Subnet boundaries support fault isolation; summarization alone does not isolate faults.

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ADDRESS RANGE REFERENCE

SPECIAL-PURPOSE AND PRIVATE RANGES

Prefix or range	Purpose	Practical meaning
10.0.0.0/8	RFC 1918 private space	For internal networks; not globally routed. Public Internet access commonly uses Network Address Translation (NAT).
172.16.0.0/12		
192.168.0.0/16		
127.0.0.0/8	Loopback	Local host; commonly 127.0.0.1.
169.254.0.0/16	Link-local	Local-link communication; not router-forwarded.
100.64.0.0/10	Shared / CGNAT	Carrier-Grade NAT space; not RFC 1918.
192.0.2.0/24 · 198.51.100.0/24 203.0.113.0/24	Documentation	Use in examples, not production addressing.
0.0.0.0/0	Default route	Matches every destination; used when no more-specific route matches.

172.16/12 covers 172.16.0.0–172.31.255.255, not all of 172/8.

Do not confuse default route 0.0.0.0/0 with the unspecified address 0.0.0.0.

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